

Mini mock

Green Machines + Metallic materials · Year 9 Science

17 minutes · each question: A / M / E

Answer every question. Use full sentences for every **explain** question.

1 Spelling counts in this question. **[A]**

- (a) Name the tubes that carry glucose around a plant. _____
- (b) Name the sticky part of a flower where pollen lands. _____
- (c) Name the property of a metal that can be stretched into wires. _____

2 Explain why pea seeds need oxygen to germinate. **[M]**

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3 A piece of metal has a mass of 53.4 g. When it is put into a measuring cylinder, the water rises from 15.0 mL to 21.0 mL. **[A / M]**

- (a) Calculate the volume of the metal.
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- (b) Calculate the density of the metal. Give the unit.
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4 Two leaves on the same plant are used. Leaf 1 is covered on both sides with foil. Leaf 2 is not covered. After 2 days in sunlight, both leaves are tested with iodine. **[E]**

- (a) Predict the colour of each leaf after the iodine test.
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- (b) Explain your predictions.
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- (c) Why was Leaf 2 needed in this experiment?
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5 Explain why kitchen sinks are made of stainless steel and not ordinary iron. **[M]**

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6 Tin reacts slowly with dilute hydrochloric acid. Copper does not react with it. **[E]**

(a) Write the word equation for tin reacting with dilute hydrochloric acid.

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(b) Copper is used for water pipes, but iron water pipes need to be protected. Explain why, using the reactivity series.

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